

Sensors are not modalities: recognition as a basal result of an embodied modulatory network

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Abstract

Recognising an object—telling it from others—is treated in contemporary AI as a pattern-classification problem solved by training a neural network on a labelled corpus. We show that recognition needs no mechanism beyond the basal architecture of an embodied agent: the same located, opponent-modulated zones that let a Sensation Modulating Network (SMN) forage also *individuate* an object, actively and with no labelled data. In a physics-based bench an agent approaches an object and reads three modalities—touch (whisker angular extent), vision (rendered luminance), and a chemical “taste” field. The number of objects it can individuate equals $2^{(\text{modalities coupled through modulation})}$: coupling more modalities multiplies the categories, but adding more transducers of one modality does not—*sensors are not modalities*. Discrimination accuracy rises with sensor count only when modulation integrates the channels; without it, more sensors add nothing—a prediction a trained classifier would bet against. Locating the object in the agent’s own frame requires refference, the self/world split. Against the identical task a subsumption controller (suppression rather than modulation) individuates at most two categories, and a corpus-trained network needs ~ 128 labelled examples to match what the SMN architecture achieves zero-shot. Recognition, on this account, scales with *modulatory architecture*—not with transducer richness or a training corpus. All results are reproducible from an open bench.

1 Introduction

The dominant route to machine recognition is to train a neural network on a labelled corpus until it maps inputs to class labels. The Sensation Modulating Network [9] suggests a different account: cognition is what a network of coupled dynamical systems—an embodied agent of sensing, modulating zones—*does* as it engages its world. A natural first reaction is to pitch this as “pattern recognition can be performed by a network of dynamical systems instead of a neural network.” That pitch is a trap. It is already established that recurrent dynamical systems classify—reservoir computing [7, 8], neural ODEs [4], liquid time-constant networks [6], and the minimally-cognitive agents of Beer [1] all recognise as dynamical systems. Restating that would be neither new nor a contribution, and it would fight on the classifier’s home ground (static input, fixed corpus), discarding what an embodied architecture actually brings.

We make instead a positive, falsifiable claim about *what* recognition costs in such an architecture. Three results, all measured on the same bench:

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1. **Recognition is not a separate competence.** The same located, opponent-modulated zones that forage in earlier experiments individuate an object—no new module, no new primitive—and they do it *actively*, driving up to the object and reading it, with no labelled corpus and no backpropagation.
2. **Sensors are not modalities.** What individuates an object is the *coupling of distinct modalities through modulation*, not transducer richness within one. The number of individuable categories is $2^{(\text{modalities coupled})}$; adding more whiskers (more of one transducer) does not add categories, and discrimination accuracy rises with sensor count only where modulation integrates the channels. This is the prediction a trained classifier—for which more input channels are simply more features—would bet against.
3. **The architecture stands in for the corpus.** On the identical task, a subsumption controller (fixed-priority suppression rather than modulation) individuates at most two categories; a corpus-trained network needs ~ 128 labelled examples to reach what the SMN architecture achieves zero-shot. And locating the object in the agent’s own frame requires reafference—the self/world split.

The contribution is therefore not “dynamical systems can recognise too,” but *recognition scales with modulatory architecture, not with transducer count or a labelled corpus*—and a concrete, reproducible demonstration of it.

2 Related work

Recognition by dynamical systems. That a recurrent dynamical system can categorise is not in question. Reservoir computing reads a class off the trajectory of a perturbed dynamical reservoir [7, 8]; neural ODEs and liquid time-constant networks are classifiers that are literally continuous-time dynamical systems [4, 6]. Closest to us, Beer [1] evolved continuous-time recurrent agents that perform *active categorical perception*—catching some shapes, rejecting others. Our point of departure from this lineage is specific: we do not evolve or train a network to the task, and our discriminating mechanism is the *modulatory coupling of multiple modalities* (with opponent zones and a self/world split), not a single recurrent substrate optimised end-to-end.

Embodiment without a dataset. Brooks [3] argued for intelligence without representation: layered reactive behaviours, no labelled world model. The SMN keeps that no-dataset, embodied commitment but replaces *suppression* (a higher layer overriding a lower one) with *modulation* (graded, antagonistic coupling that binds channels), and it does build a constructed, body-relative snapshot. Subsumption historically hit a ceiling at higher cognition; we show below, as a measured arm, exactly where suppression stalls. Pfeifer and Bongard [11] on morphological computation is an ally: much of what looks like computation is paid for by the body and its coupling—here, by the modulatory architecture and the cross-modal convergence it affords. Our reading of perception as constituted in sensorimotor engagement follows the enactive/sensorimotor tradition [10] and Gibson’s affordances [5]; interactive perception in robotics [2] similarly leverages action, typically still atop learned perception.

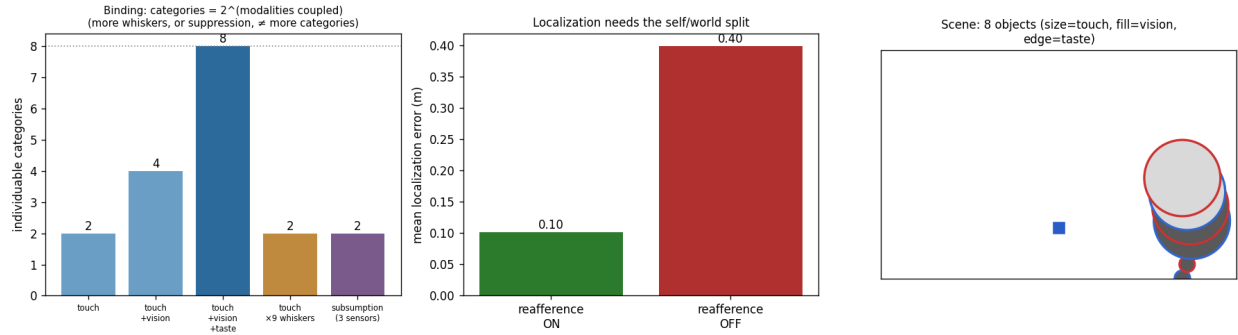


Figure 1: **Binding and localisation.** *Left:* individuatable categories = $2^{(\text{modalities coupled})}$ ($2 \rightarrow 4 \rightarrow 8$); a touch-only agent with nine whiskers still individuates only two, and a subsumption controller with all three sensors also only two—neither transducer density nor suppression adds categories. *Centre:* mean localisation error is $\sim 5\times$ worse without refference, and approximates the distance the agent travelled—the signature of attributing self-motion to the world. *Right:* the eight objects (size=touch, fill=vision, edge=taste).

3 The bench and the cross-modal architecture

The experiments run on **smn-lab**, an open MuJoCo bench in which the SMN control architecture drives a physical body [9]. A planar agent (a “mouse”) has translational and yaw degrees of freedom, a fan of rangefinder whiskers, a forward-facing camera, and an inertial site for proprioception. Locomotion is produced by a basal action pattern (a central pattern generator) and two *located* opponent drive zones whose net force and torque follow from the body geometry—so the controller that forages in prior experiments is, unchanged, the controller that approaches here.

The object to be recognised carries one feature bit per modality (Figure 1, right): its **radius** sets the tactile bit (a bigger object subtends more whiskers at a fixed gap); its **luminance** sets the visual bit, read from the rendered camera image pooled into patch tokens; and a Gaussian chemical field at the object sets the **taste** bit, sampled where the agent stands (so it must approach to taste). With three bits there are eight object types.

The independent variable is the *cross-modal board*—the experiment’s “balance beam.” It (a) *modulates*: each modality’s reading is gated against a noise floor, so only a reading that breaks the floor flows; and (b) *couples*: the surviving bits of the wired-in modalities are concatenated into one object code. The visual channel is surfaced by *refference*: at closest approach a small forward probe makes the object loom in a way the agent’s rotation-only forward model cannot predict, and the modulator gates the unpredicted change—figure-ground from the agent’s own motion. Localisation places the object in the agent’s dead-reckoned frame, built from proprioception rather than any absolute pose.

4 Results

We report four measurements. Binding, localisation, and the foils run on the embodied bench (one object per trial; the agent approaches and reads at closest approach). The resolution sweep is a controlled Monte-Carlo isolating the integration principle. All randomness is seeded; the experiment and figures are regenerable from a single script.

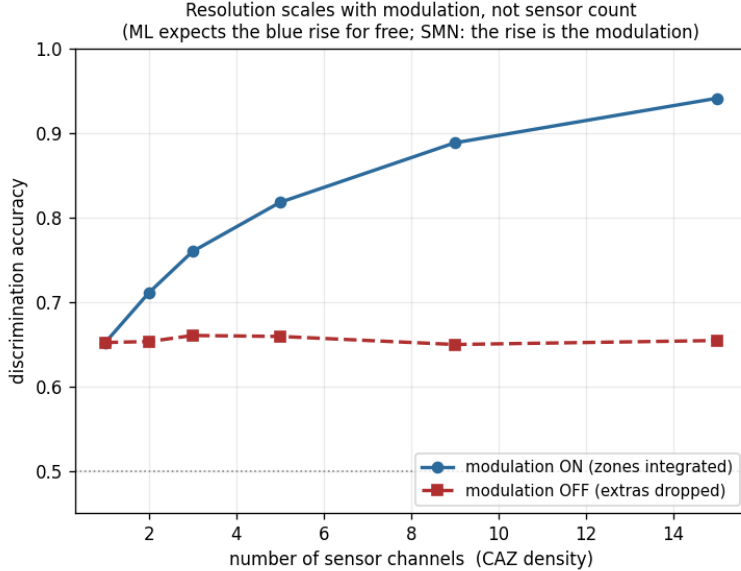


Figure 2: **Resolution scales with modulation, not sensor count.** With N redundant noisy channels, modulation integrates the gated zones and accuracy rises as $\sigma/\sqrt{N}$ ($0.65 \rightarrow 0.94$); strip the modulation and the ungated extras are dropped, so more sensors are flat (≈ 0.65). A trained classifier expects the rising curve from raw channels for free; the SMN claim is that the rise is bought by the modulatory architecture.

4.1 Binding: the object is a cross-modal invariant

Decoding all eight objects with the cross-modal board, the number of individuable categories equals 2 raised to the number of modalities coupled through modulation: **2** (touch) $\rightarrow$ **4** (touch+vision) $\rightarrow$ **8** (touch+vision+taste), with every resolved bit correct (Figure 1, left). One modality collapses the set fourfold; coupling individuates. Crucially, giving the touch-only agent *nine* whiskers instead of five leaves it at **2** categories: a touch-only agent cannot read colour no matter how many whiskers it grows. The empirical content is the dissociation, not the multiplication—*sensors are not modalities*.

4.2 The resolution principle: modulation, not sensor count

Why does density within a modality not help? Because resolution is a function of the modulatory architecture, not of the raw transducer. In a controlled two-class discrimination from N redundant noisy channels (Figure 2), *with* modulation the modulator integrates the gated zones and accuracy climbs from 0.65 ($N=1$) to 0.94 ($N=15$) as $\sigma/\sqrt{N}$; *without* modulation the ungated extras are dropped and accuracy is flat (≈ 0.65) regardless of N . The rise is not a property of having more sensors—it is bought by the architecture that integrates them. This is precisely the prediction a feature-hungry classifier would bet against.

4.3 Localisation needs refference

The agent reports not just *what* the object is but *where*, in its own dead-reckoned frame—so the world model includes locating objects in space. Holding the object at a stable place while the agent moves toward it requires discounting self-caused sensory change. With refference on, mean localisation error is ~ 0.10 m; with it off, ~ 0.40 m—roughly the distance the agent travelled, because

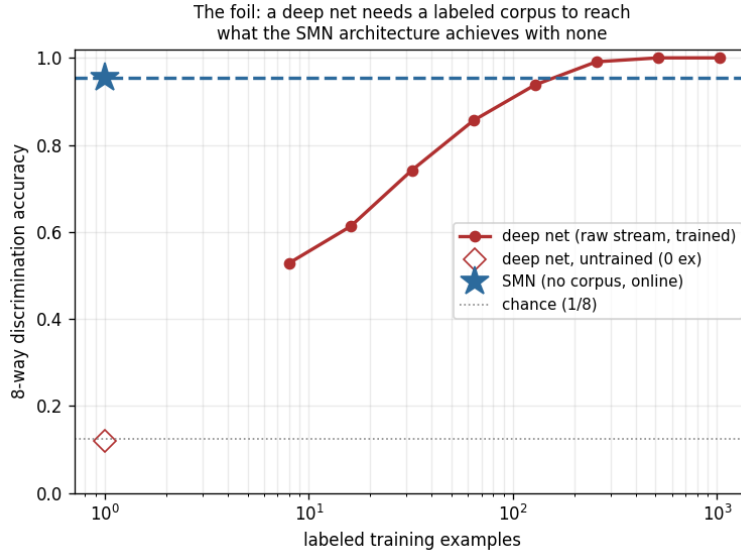


Figure 3: **The architecture stands in for the corpus.** On the same generative test set, an untrained network sits at chance (~ 0.12) and the SMN architecture scores ~ 0.96 with *zero* labelled examples; a trained network needs ~ 128 labelled examples to match it. At the same zero corpus the gap is the whole distance from chance to near-perfect.

System (all three sensors)	Labels	Categories / accuracy
SMN (cross-modal board)	0	8 (~ 0.96 on the noisy set)
Subsumption (suppression)	0	2
Deep net (untrained)	0	~ 0.12 (chance)
Deep net (trained)	~ 128	~ 0.96 (matches SMN)

Table 1: The same task, three systems. Modulation binds (8); suppression does not (2); a trained network matches SMN only after a corpus.

the agent then attributes its own approach to the object (Figure 1, centre). The self/world split is not an add-on to recognition; it is what makes the located object stable.

4.4 Foils: the architecture against suppression and against a corpus

We compare the SMN architecture to its two natural foils on the identical task (Table 1). The *subsumption* foil shares the body, the three sensors, and the reactive approach, but arbitrates by fixed-priority *suppression* rather than modulation: the highest-priority firing layer suppresses the rest and emits one bit. Because suppression replaces rather than binds, it individuates at most **2** categories with all three sensors present—the ceiling that modulation breaks through to 8. The *deep-net* foil is a small network trained on the raw sensory stream (whisker ranges, camera tokens, taste). On a generative test set drawn from the same per-object readings plus channel noise, an untrained network sits at chance (~ 0.12 , $1/8$) and a trained one reaches the SMN architecture’s zero-shot accuracy (~ 0.96) only after ~ 128 labelled examples (Figure 3). What a network must learn from a corpus, the SMN architecture supplies as an embodied prior.

5 Discussion

Across the four measurements a single point recurs: *the modulatory architecture is doing the work that classifiers buy with data*. Recognition is not a competence layered on top of the agent—it is what the basal, located, opponent-modulated zones already do when coupled across modalities. The recognised category is the equivalence class of scenes that drive the same coupled code; the object is the cross-modal invariant, determinate where distinct modalities, coupled through modulation, agree. Resolution is architectural, not a function of transducer count; and locating the object requires the same self/world split the agent uses to map. None of this needs a new mechanism or a labelled corpus.

The account is deliberately bounded. *Naming*—mapping an object code to an utterance—is another modulatory coupling, but a name is conventional and social: it requires at least a dyad, and is the subject of the next monograph in this programme rather than of this paper. The encounter here is staged one object at a time for a controlled read; recognition emerging during open foraging is a natural next step. And the refference gate uses a rotation-only forward model, so it surfaces the object from self-caused translation it cannot yet predict—a figure-ground signal, not yet a model that predicts looming. The strength of the result is what it refuses to claim: a single, falsifiable dissociation (sensors $\neq$ modalities; architecture $\neq$ corpus) demonstrated end to end.

6 Reproducibility

The bench, the experiment, and an interactive lab interface are open (github.com/gnowgi/smn-lab, GPL-3.0). Every result above is produced by a single seeded script (`experiments/p3_crossmodal_discrimination.py`); the three figures regenerate from it. A Streamlit interface lets a reviewer run the controller live—watch the agent approach and read each object, and compute the result tables on the spot—so the figures are visibly the math in action rather than reported numbers. A versioned snapshot with a Zenodo DOI will accompany the published version.

7 Conclusion

Recognition, in an embodied Sensation Modulating Network, is a basal result: the same modulatory zones that keep the agent alive individuate an object, actively and without a corpus. What individuates is the coupling of modalities through modulation, and what makes the object’s place stable is refference. The clear, contestable claim—*recognition scales with modulatory architecture, not with transducer richness or a labelled corpus*—is borne out by a measured dissociation against both suppression and a trained network. Sensors are not modalities; architecture is not a corpus.

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